

JEE MAIN REVISION MASTER: LOGARITHMS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 2 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Standard	Logarithm — Core Principles & Governing Laws	1 PYQs	Core Principle
Standard	Logarithm — Standard Calculations & Problem Archetypes	1 PYQs	Quantitative Analysis
Standard	Logarithm — Applied Cases, Reactions & Traps	0 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Logarithm — Core Principles & Governing Laws	$\ln R^{+}$
Logarithm — Standard Calculations & Problem Archetypes	$e^0=1$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Logarithm — Core Principles & Governing Laws (Standard Yield — 1 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Logarithm.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Logarithm.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Logarithm PYQs.

Representative Source PYQs:

• [2024 Shift] Q#1 (MCQ): "Which one of the following observations is correct for the features of logarithm function to any base $b>1$?..." [Ans: N/A]

3.2 Logarithm — Standard Calculations & Problem Archetypes (Standard Yield — 1 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Logarithm.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2023 Shift] Q#2 (MCQ): "The value of $e^{\log_{10} \tan 1^\circ + \log_{10} \tan 2^\circ + \log_{10} \tan 3^\circ + \dots + \log_{10} \tan 89^\circ}$ is..." [Ans: N/A]

3.3 Logarithm — Applied Cases, Reactions & Traps (Standard Yield — 0 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Logarithm.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2024 Shift] Q#1 (MCQ): "Which one of the following observations is correct for the features of logarithm function to any base $b > 1$?..." [Ans: N/A]
- [2023 Shift] Q#2 (MCQ): "The value of $e^{\log_{10} \tan 1^\circ + \log_{10} \tan 2^\circ + \log_{10} \tan 3^\circ + \dots + \log_{10} \tan 89^\circ}$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2024 (Q#1)

Which one of the following observations is correct for the features of logarithm function to any base $b > 1$?

- (A) The domain of the logarithm function is $\mathbb{R}$, the set of real numbers.
- (B) The range of the logarithm function is $\mathbb{R}^+$, the set of all positive real numbers.
- (C) The point (1,0) is always on the graph of the logarithm function.
- (D) The graph of the logarithm function is decreasing as we move from left to right.

Correct Answer: None

Solution Analysis: ■ Domain $\in \mathbb{R}^+$ and range $\in \mathbb{R}$ and graph of the logarithm function is increasing. So, point (1,0) is always on the graph of logarithm function.

Benchmark Problem 2: KCET 2023 (Q#2)

The value of $e^{\log_{10}\tan 1^\circ + \log_{10}\tan 2^\circ + \log_{10}\tan 3^\circ + \dots + \log_{10}\tan 89^\circ}$ is

- (A) 3
- (B) $(1/e)$
- (C) 1
- (D) 0

Correct Answer: None

Solution Analysis: We know that $\tan\theta \tan(90^\circ - \theta) = \tan\theta \cdot \cot\theta = 1$ $\tan 1^\circ \tan 89^\circ = \tan 2^\circ \tan 88^\circ = \tan 3^\circ \tan 87^\circ = \tan 44^\circ \tan 46^\circ = 1$

So, $\log_{10}\tan 1^\circ + \log_{10}\tan 2^\circ + \dots + \log_{10}\tan 89^\circ$

$= (\log_{10}\tan 1^\circ + \log_{10}\tan 89^\circ) + (\log_{10}\tan 2^\circ + \log_{10}\tan 88^\circ) + \dots + (\log_{10}\tan 44^\circ + \log_{10}\tan 46^\circ) + \log_{10}\tan 45^\circ$
 $= \log_{10}(\tan 1^\circ \tan 89^\circ) + \log_{10}(\tan 2^\circ \tan 88^\circ) + \dots + \log_{10}(\tan 44^\circ \tan 46^\circ) + \log_{10}\tan 45^\circ$
 $= \log_{10}1 + \dots + \log_{10}1 + \log_{10}1$